Yes. Since Loans now already has **Defaulted (Yes/No)** and **Default Loss Amount**, I would make **no AppSheet Database schema changes** for Page 7.

Below is a clean implementation specification you can hand directly to Codex. The intended architecture is:

Existing database tables stay unchanged.
Add only AppSheet virtual columns/calculations in the **OLAP app**, then use Looker Studio parameters/calculated fields for the what-if assessment.

Page 7 Specification — Borrower Insight & New Loan Assessment

1. Objective

Create a single-borrower analytical page in Looker Studio to support a **new-loan decision**.

The page should answer:

1. What is the borrower's current exposure?
2. How reliably has she paid historically?
3. Is her recent payment behaviour improving or deteriorating?
4. What has been her proven historical borrowing capacity?
5. Has she defaulted before, and how severe was the loss?
6. What would happen to borrower exposure and portfolio concentration if a proposed new loan is granted?

Do **not** implement an opaque numerical credit/risk score.

The page should expose the underlying risk indicators so the lending decision remains explainable.

---

2. Architecture

Use the following sources:

Data	Main source
Borrower-level current metrics	Borrowers
Loan history / defaults	Loans

Data	Main source
Payment behaviour	Charges
Profit	Repayments
Portfolio totals	Borrower VC / existing Statistics
Proposed new loan	Looker Studio parameter

No new physical database table or column is required.

Any additional AppSheet fields below should be **VCs in the OLAP app only**.

---

3. AppSheet changes

3.1 Charges — add Days Late VC

Purpose: common building block for overdue history.

Type:

Number

Logic:

- if charge was fully paid, calculate number of days between Charge Date and final Payment Date;
- minimum is 0;
- if the charge remains unpaid and Charge Date is before today, calculate TODAY() - Charge Date;
- today's unpaid charge is not yet late.

Conceptual expression:

```
IFS(
 AND(
  [Payment Status] = "ชำระแล้ว",
  ISNOTBLANK([Payment Date])
 ),
 MAX(
  LIST(
```

```
    0,
    [Payment Date] - [Charge Date]
  )
 ),

 [Charge Date] < TODAY(),
 TODAY() - [Charge Date],

 TRUE,
 0
)
```

Use the existing Thai fully-paid value already used in Payment Status.

3.2 Charges — add Historical On Time VC

Type:

Yes/No

Purpose:

Identify charges that were actually settled on or before their due/charge date.

```
AND(
 [Charge Date] < TODAY(),
 [Payment Status] = "ชำระแล้ว",
 [Days Late] = 0
)
```

Today's charge is deliberately excluded so that an unpaid charge at lunchtime does not damage the borrower's historical reliability rate.

3.3 Charges — add Historical Late VC

Type:

Yes/No

AND(

[Charge Date] < TODAY(),

[Days Late] > 0

)

This includes both:

- charges eventually paid late;
- currently overdue historical charges.

---

4. Borrowers — payment-behaviour VCs

Use the existing relationship:

Borrowers

↓

Loans

↓

Charges

with the established relationship path:

[Ref Loans].[Ref Borrowers]

Adjust only the exact borrower key reference if necessary.

4.1 Historical Charge Count

Type:

Number

Count charges with:

Charge Date < TODAY()

for the borrower.

Conceptually:

COUNT(

```
 SELECT(
  Charges[Row ID],
  AND(
   [Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID],
   [Charge Date] < TODAY()
  )
 )
)
```

4.2 On Time Charge Count

```
COUNT(
 SELECT(
  Charges[Row ID],
  AND(
   [Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID],
   [Historical On Time]
  )
 )
)
```

4.3 Lifetime On Time Rate

Type:

Percent

```
IF(
 [Historical Charge Count] = 0,
 0,
 [On Time Charge Count]
 /
```

```
 [Historical Charge Count]
)
```

4.4 Recent 30D Charge Count

Use completed historical due dates only:

TODAY()-30 <= Charge Date < TODAY()

Conceptually:

```
COUNT(
 SELECT(
  Charges[Row ID],
  AND(
   [Ref Loans].[Ref Borrowers] = [_THISROW].[Row ID],
   [Charge Date] >= TODAY() - 30,
   [Charge Date] < TODAY()
  )
 )
)
```

4.5 Recent 30D On Time Count

Same period plus:

Historical On Time = TRUE

4.6 Recent 30D On Time Rate

```
IF(
 [Recent 30D Charge Count] = 0,
 0,
 [Recent 30D On Time Count]
 /
```

[Recent 30D Charge Count]

)

This becomes one of the most important Page 7 indicators.

4.7 Lifetime Late Payment Count

Count:

Historical Late = TRUE

4.8 Recent 30D Late Payment Count

Count historical late charges where:

Charge Date >= TODAY()-30

AND

Charge Date < TODAY()

4.9 Maximum Days Late

Type:

Number

Maximum Charges[Days Late] for this borrower.

Return 0 if no late history exists.

4.10 Last Late Payment Date

Type:

Date

Return the most recent:

Charge Date

where:

Days Late > 0

4.11 Current Overdue Amount

Use your existing Charges[Amount Remaining].

Sum where:

Charge Date < TODAY()

AND

Amount Remaining > 0

This is different from today's unpaid collection.

Today's unpaid charge must not count as overdue.

---

5. Borrowers — loan-history VCs

These metrics should use Loans.

5.1 Active Loan Count

Count loans belonging to the borrower that are currently active.

Use the existing loan-status logic rather than introducing new status values.

5.2 Current Outstanding Principal

Reuse the existing borrower-level outstanding-principal calculation if already present.

Do **not** create another duplicate VC if one already exists.

5.3 Completed Loan Count

Count closed loans belonging to the borrower.

This includes defaulted loans unless specifically labelled otherwise.

5.4 Successfully Completed Loan Count

Count loans where:

Closed

AND

Defaulted = FALSE

This is preferable for determining demonstrated successful borrowing history.

5.5 Largest Successful Loan

Type:

Price

Return:

MAX(Loans[Principal Amount])

where:

Borrower = current borrower

Closed

Defaulted = FALSE

This is one of the most important new-loan assessment fields.

5.6 Average Successful Loan Size

Average Principal Amount for:

Closed

AND

Defaulted = FALSE

Do not include current active loans.

The purpose is to establish the borrower's **demonstrated historical borrowing range**.

5.7 Default Count

Count loans where:

Defaulted = TRUE

Because Defaulted is already physical, this metric is now reliable.

5.8 Lifetime Default Loss

SUM(Default Loss Amount)

for the borrower.

5.9 Largest Default Loss

Optional but inexpensive VC:

MAX(Default Loss Amount)

for defaulted loans.

Useful in the overdue/default panel, but not mandatory for the first version.

---

6. Borrowers — portfolio context

6.1 Portfolio Outstanding Principal

Add one OLAP-only borrower VC returning the **total current portfolio outstanding**, regardless of borrower.

Conceptually:

SUM(

  Loans[Outstanding Principal]

)

subject to your existing active/outstanding logic.

Every borrower row intentionally receives the same number.

Why duplicate it as a VC?

Because when Looker Studio filters Page 7 to one borrower, the denominator must still represent the **entire portfolio**.

If Looker derives both numerator and denominator from the filtered borrower data, portfolio concentration can incorrectly become 100%.

This VC avoids that issue cleanly without adding any database column.

6.2 Current Portfolio Share

Borrower VC:

IF(

[Portfolio Outstanding Principal] = 0,

0,

[Current Outstanding Principal]

/

[Portfolio Outstanding Principal]

)

Format:

Percent

---

7. Profit metrics

Reuse existing borrower VCs wherever possible.

At minimum Page 7 needs:

Lifetime Profit

Optionally:

Profit Last 30 Days

If these already exist in Borrowers, reuse them.

If not, Page 7 can calculate lifetime profit from Repayments, but for a single-borrower dashboard I would prefer an existing borrower-level OLAP VC over a Looker blend.

Do **not** create physical database fields.

---

8. Looker Studio parameter

Create:

Proposed Loan Amount

Type:

Number

Default:

0

Suggested control:

Input box

Display label:

Proposed New Loan

Format visually as THB where possible.

---

9. Looker calculated fields

9.1 Projected Borrower Exposure

Current Outstanding Principal

+

Proposed Loan Amount

9.2 Projected Portfolio Outstanding

Portfolio Outstanding Principal

+

Proposed Loan Amount

9.3 Projected Portfolio Share

Projected Borrower Exposure

/

Projected Portfolio Outstanding

Format:

Percent

9.4 Exposure vs Historical Maximum

Useful contextual ratio:

Projected Borrower Exposure

/

Largest Successful Loan

For example:

2.1x historical maximum

Do not interpret this automatically as good or bad yet.

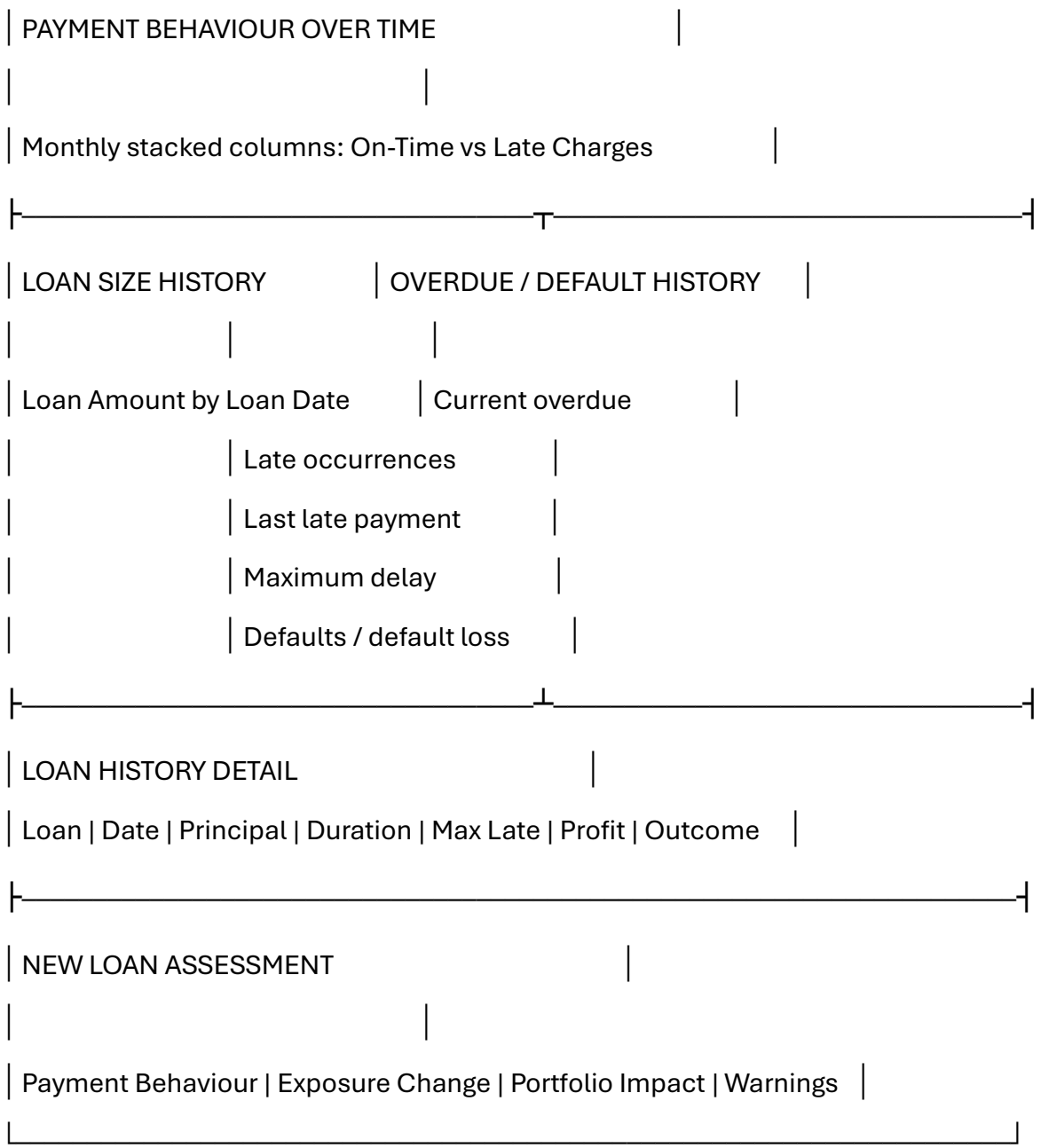
It is supporting evidence.

---

10. Final Page 7 visual layout

Use this approximate structure:

```
┌──────────────────────────────────────────────────────────────┐
│ BORROWER INSIGHT & NEW LOAN ASSESSMENT                       │
│ [Borrower ▼]                 Proposed Loan [ ฿_____ ]        │
├──────────────────────────────────────────────────────────────┤
│ CURRENT POSITION                                             │
│                                                              │
│ Outstanding     Active Loans     Current Overdue   Lifetime  │
│ Principal                                         Profit     │
│ ฿15,000         2                ฿0               ฿18,450    │
├──────────────────────────────┬───────────────────────────────┤
│ EXPOSURE ANALYSIS            │ PAYMENT RELIABILITY           │
│                              │                               │
│ Current Exposure             │ Lifetime On-Time      94%     │
│ Proposed Loan                │ Recent 30D On-Time    82%     │
│ Projected Exposure           │ Late Payments         4       │
│                              │ Recent Late Payments  2       │
│ Historical Avg / Maximum     │ Maximum Delay         5 days  │
│                              │ Defaults              0       │
├──────────────────────────────┴───────────────────────────────┤
```



---

11. Page control — Borrower selector

Use a single-select borrower control.

Field:

Borrower Name

Set:

Single select = ON

Page 7 should always be interpreted as a **single-borrower page**.

Do not encourage multi-select.

All charts and scorecards should respond to this borrower control except portfolio-total denominators where deliberately preserved.

---

12. Row 1 — Current Position

Use four scorecards.

Scorecard 1 — Current Outstanding Principal

Source:

Borrowers

Metric:

Current Outstanding Principal

Scorecard 2 — Active Loans

Metric:

Active Loan Count

Scorecard 3 — Current Overdue

Metric:

Current Overdue Amount

This should show:

฿0

when there is no overdue amount rather than blank.

Scorecard 4 — Lifetime Profit

Metric:

existing:

Lifetime Profit

These four cards establish:

Exposure + complexity + immediate warning + relationship value.

---

13. Exposure Analysis panel

This should occupy approximately 50% of the next row.

Visualization 1 — Current vs Projected Exposure

Preferred chart:

Horizontal bar

Dimension can use small calculated/static categories or separate scorecard-like elements.

Show:

Current Outstanding ฿15,000

Projected Exposure ฿25,000

Largest Successful Loan ฿12,000

Average Successful Loan ฿6,500

The visual purpose is to compare the proposed exposure against the borrower's proven borrowing history.

Do not use a gauge.

A comparison bar is much more informative.

---

14. Visualization 2 — Portfolio Concentration Before / After

Place directly underneath or beside the exposure comparison.

Show:

Current Portfolio Share 16%

Projected Portfolio Share 24%

Use either:

- two horizontal bars; or
- two large comparison values.

This answers:

Even if the borrower is reliable, would the new loan create excessive concentration?

---

15. Payment Reliability panel

Use compact KPI/value blocks rather than too many charts.

Show:

Lifetime On-Time Rate

94%

Recent 30D On-Time Rate

82%

Lifetime Late Payments

4

Late Payments – Last 30D

2

Maximum Days Late

5

Defaults

0

The two on-time percentages should be visually paired.

The page should make situations like this obvious:

Lifetime 96%

Recent 72%

because recent deterioration is highly relevant to the new-loan decision.

---

16. Visualization 3 — Payment Behaviour Over Time

Use:

100% stacked column chart or normal stacked column chart.

Recommended first version:

Normal stacked column

Dimension:

Charge Date → Year Month

Series:

On-Time Charges

Late Charges

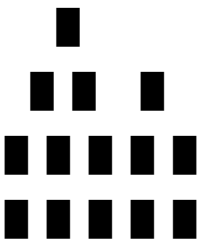
Metric:

Record Count

Purpose:

Show whether late payments are becoming more frequent.

Example:



Apr May Jun Jul Aug

Because Page 7 is filtered to one borrower, no borrower dimension is needed.

---

17. Visualization 4 — Loan Amount History

Use:

Column chart

Data source:

Loans

Dimension:

Loan Date

Metric:

Principal Amount

One bar = one loan.

Sort:

Loan Date ascending

Do not aggregate multiple loans occurring on different dates into monthly totals if the individual-loan history can be shown clearly.

The purpose is to identify patterns like:

฿3k → ฿5k → ฿5k → ฿8k → ฿15k

which shows how exposure has progressively increased.

---

18. Overdue / Default History panel

Use compact metrics rather than another complicated chart.

Show:

Current Overdue

Current Overdue Amount

Late-payment occurrences

Lifetime Late Payment Count

Late in last 30 days

Recent 30D Late Payment Count

Maximum delay

Maximum Days Late

Last late payment

Last Late Payment Date

Previous defaults

Default Count

Lifetime default loss

Lifetime Default Loss

If Default Count = 0, default loss should naturally be ฿0.

---

19. Loan History Detail table

Full-width table.

Data source:

Loans

Recommended fields:

Column	Source
Loan ID	Loans
Loan Date	Loans
Principal	Loans
Loan Duration	Loans
Status	Loans
Defaulted	Loans
Default Loss Amount	Loans
Profit	existing loan-level calculation if available

Ideally also include:

Maximum Days Late for Loan

If this is not currently available, it can be a **Loans VC**, derived from related Charges.

---

20. Optional Loans VC — Maximum Days Late

This one is worth adding because it significantly improves the loan-history table and requires no database change.

Conceptually:

```
MAX(
 SELECT(
  Charges[Days Late],
  [Ref Loans] = [_THISROW].[Loan ID]
 )
)
```

Return 0 when there are no matching late charges.

Then table becomes:

Loan Principal Duration Max Late Defaulted Profit

That is an excellent historical behavioural summary.

---

21. Outcome display field

In Looker, create a display-only calculated dimension:

```
CASE
 WHEN Defaulted = TRUE THEN "Defaulted"
 WHEN Status = Closed THEN "Repaid"
 ELSE "Active"
END
```

Do **not** add another database column for this.

Your physical truth remains:

Status

Defaulted

The Looker field merely improves presentation.

---

22. New Loan Assessment section

I would **not implement one black-box risk score**.

Instead show four assessment dimensions.

A. Payment Reliability

Base it primarily on:

Lifetime On-Time Rate

Recent 30D On-Time Rate

Current Overdue

Default Count

Display the raw values prominently.

B. Exposure Change

Show:

Current Outstanding

Proposed Loan

Projected Exposure

Largest Successful Loan

Example:

Current ฿15,000

New ฿10,000

Projected ฿25,000

Historical maximum ฿12,000

C. Portfolio Impact

Show:

Current Portfolio Share

Projected Portfolio Share

Example:

16% → 24%

D. Warning Indicators

Use a compact table/list generated from transparent conditions.

Examples:

Recent on-time rate lower than lifetime rate

Current overdue exists

Previous default exists

Projected exposure exceeds historical maximum

Projected portfolio concentration materially increases

Do not write these warnings into the database.

They should be Looker-calculated presentation logic.

---

23. Suggested initial risk-status logic

Codex should implement these as **Looker calculated fields only**, so thresholds remain easy to change.

They are intended as starting presentation rules, not permanent credit policy.

Payment Behaviour Status

High Concern

If:

Default Count > 0

OR

Current Overdue Amount > 0

Review

Otherwise if:

Recent 30D On-Time Rate < 90%

Good

Otherwise.

Exposure Status

High

If:

Projected Exposure

>

2 × Largest Successful Loan

Elevated

If:

Projected Exposure

>

Largest Successful Loan

Within History

Otherwise.

Special handling:

If there is **no successful historical loan**, display:

No established history

rather than treating the denominator as zero.

Concentration Status

Do not hard-code a permanent policy threshold yet.

For initial visualization only:

Projected Share <= 25% Normal

25–40% Elevated

>40% High

These must remain easy to edit.

---

24. Overall assessment

I would actually make the initial Page 7 **not calculate a numeric overall score**.

Instead display something like:

Payment Behaviour GOOD

Recent Behaviour REVIEW

Exposure ELEVATED

Portfolio Concentration NORMAL

Then the person making the loan decision sees *why* there is concern.

If you later want an overall:

LOW / MODERATE / HIGH

risk classification, add it after you have enough actual default/payment history to calibrate the rules.

That is much more defensible than inventing a score now.

---

25. AppSheet changes summary for Codex

Charges — add VCs

Days Late

Historical On Time

Historical Late

Borrowers — add/reuse VCs

Historical Charge Count

On Time Charge Count

Lifetime On Time Rate

Recent 30D Charge Count

Recent 30D On Time Count

Recent 30D On Time Rate

Lifetime Late Payment Count

Recent 30D Late Payment Count

Maximum Days Late

Last Late Payment Date

Current Overdue Amount

Active Loan Count

Current Outstanding Principal ← reuse if existing

Completed Loan Count

Successfully Completed Loan Count

Largest Successful Loan

Average Successful Loan Size

Default Count

Lifetime Default Loss

Portfolio Outstanding Principal

Current Portfolio Share

Lifetime Profit ← reuse if existing

Loans — add one optional but recommended VC

Maximum Days Late

Physical database changes

NONE

Existing physical fields:

Loans[Defaulted]

Loans[Default Loss Amount]

are sufficient for default history.

---

26. Looker components summary

The finished page should contain:

#	Component	Source
1	Borrower selector	Borrowers
2	Proposed Loan parameter	Looker
3	Current Outstanding	Borrowers
4	Active Loans	Borrowers
5	Current Overdue	Borrowers
6	Lifetime Profit	Borrowers
7	Current vs Projected Exposure	Borrowers + parameter
8	Historical Avg / Max Loan comparison	Borrowers
9	Portfolio Share Before / After	Borrowers + parameter
10	Lifetime vs Recent On-Time	Borrowers
11	Payment Behaviour by Month	Charges
12	Loan Amount History	Loans
13	Overdue / Default KPIs	Borrowers
14	Loan History table	Loans
15	New Loan Assessment indicators	Looker calculated fields

The key design rule for Codex is:

Do not solve Page 7 by adding reporting structures to the AppSheet Database. Enrich the existing OLAP AppSheet tables with VCs where reusable business logic is needed, and keep hypothetical/new-loan logic entirely inside Looker Studio.

That gives you the borrower underwriting view we want while keeping your underlying OLTP/OLAP database architecture essentially unchanged.